SYNTHETIC / SIMULATED DATA — GENERATED BY THE RareDiseasePipeline TEST SUITE — NOT A REAL CLINICAL RESULT, NOT FOR PATIENT CARE

RareDiseasePipeline Clinical Genome Center

Synthetic Data Validation Laboratory

Pipeline QA Environment · WSL/Colab · Nextflow DSL2
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WGS FINAL REPORT (SYNTHETIC)

Patient ID	SIM-HEMA-01	**Ordering Physician**	Dr. [Ordering Physician]	**Specimen**	Simulated peripheral blood
Sex	Male	**Account#**	SIM-PIPE	**Collected**	2026-07-15
Date of Birth	XX/XX/XXXX (synthetic)	**Source**	Pipeline v1.1 test dataset	**Received**	2026-07-15
Indication	Simulated: suspected bleeding disorder, low Factor VIII	**Case ID**	CS-HEMA_INXXXX	**Reported**	2026-07-15

TEST RESULT: A likely pathogenic missense variant and a large exon 1–2 deletion were detected in **F8** in this simulated case, and a candidate intron 22 inversion breakend was flagged pending orthogonal confirmation. Pathogenic F8 variants cause **Haemophilia A** [MIM: 306700], an X-linked bleeding disorder caused by deficient or dysfunctional coagulation Factor VIII.

PATIENT PHENOTYPE (SIMULATED)

Simulated phenotype terms used to drive prioritization: prolonged bleeding, hemarthrosis, low Factor VIII activity.

PRIMARY FINDINGS — Variants in genes associated with patient's simulated phenotype

Confirmation Status	Gene / Locus	Condition	Chrom : Coordinates	Variant	Zygosity	Inheritance	Classification
Confirmed	F8	Haemophilia A	chrX: 154,843,200	c.? p.? (missense)	Hemizygous	X-linked Recessive	Likely Pathogenic
Confirmed	F8	Haemophilia A	chrX: 154,836,200	Exon 1-2 deletion	Hemizygous	X-linked Recessive	Pathogenic
Pending Review	F8	Haemophilia A (candidate)	chrX: 154,845,200	Candidate intron 22 inversion (breakend)	Hemizygous (candidate)	X-linked Recessive	Not yet classified — requires long-read/inverse-PCR confirmation

INTERPRETATION

Variant Information

Missense variant: supported concordantly by REVEL (0.81), AlphaMissense (0.72), and CADD (27.0), all above the pipeline's pathogenicity cutoffs, as well as by ClinVar (Likely_pathogenic) and InterVar (PM1, PM2, PP3). Rare in gnomAD (AF 0.00003).

Exon 1–2 deletion: classified Pathogenic by both ClassifyCNV and SvAnna, which additionally notes loss of the start codon. F8 is haploinsufficiency-sufficient per ClinGen (HI=3), consistent with this deletion being disease-causing.

Candidate intron 22 inversion: a breakend consistent with the classic intron 22 inversion hotspot recurrently implicated in severe Haemophilia A was detected by the structural-variant caller only (1 of the available tools) and is **not present in ClinVar**. Per pipeline policy, single-tool structural events at recurrent hotspots are flagged as candidates rather than auto-classified, and require long-read sequencing or inverse-PCR for confirmation before clinical use. A common, benign intronic variant was also detected and does not affect interpretation.

Gene Information

F8 encodes coagulation Factor VIII, an essential cofactor in the intrinsic coagulation cascade. Missense variants, large deletions, and the recurrent intron 22 (and less commonly intron 1) inversions are all well-established mechanisms of severe Haemophilia A; inversions in particular are a leading cause of severe disease but are structurally complex and often missed by standard short-read SNV/indel calling, which is why this pipeline routes them through a dedicated long-read/breakend detection branch.

REFERENCES

Bagnall RD, et al. Journal of Thrombosis and Haemostasis. 2002. Recurrent inversion breaking intron 1 of the factor VIII gene is a frequent cause of severe hemophilia A.

Lakich D, et al. Nature Genetics. 1993. Inversions disrupting the factor VIII gene are a common cause of severe haemophilia A.

Richards S, et al. Genetics in Medicine. 2015. Standards and guidelines for the interpretation of sequence variants (PMID: 25741868).

RECOMMENDATIONS

- Clinical correlation with the patient's phenotype is recommended.
- Because this dataset is synthetic pipeline-validation data, no clinical action should be taken on it.
- Genetic counseling is recommended in real cases to discuss implications for the patient and family.
- Orthogonal confirmation (e.g. Sanger sequencing, long-read sequencing, or inverse-PCR where applicable) is recommended before clinical sign-out of any pathogenic or likely-pathogenic call.
- Periodic re-evaluation of variant classification is appropriate as knowledge and databases evolve.

TEST METHODOLOGY

This report was produced by the RareDiseasePipeline Nextflow/Colab workflow, a synthetic-data validation harness modeled on production rare-disease WGS interpretation pipelines. Variants shown here were generated by the pipeline's own per-disease test-data generator (Section 12.1 of the pipeline notebook), then passed through the same region-filtering, annotation, disease-specific-branch, prioritization, and ACMG classification steps used for real inputs: VEP and SnpEff for functional consequence, ClinVar/SnpSift for clinical significance, gnomAD for population frequency, SpliceAI/REVEL/AlphaMissense/CADD for in-silico pathogenicity prediction, InterVar for ACMG/AMP classification, ClassifyCNV and SvAnna for structural-variant interpretation, ClinGen for gene dosage sensitivity, and ExpansionHunter/TRGT for repeat-expansion sizing. Variant classification follows the ACMG/AMP 2015 sequence-variant interpretation guidelines (Richards et al. 2015).

TEST LIMITATIONS

This is a synthetic dataset generated for pipeline validation and does not represent a real patient or real sequencing data. As with any WGS-based test, full genome coverage is not achievable in repetitive or duplicated regions, and certain variant classes (e.g. deep intronic variants, some repeat expansions, complex structural rearrangements) may not be reliably detected. Structural-variant and inversion calls that are reported by only one tool in the chain are flagged as candidates and require orthogonal confirmation before they can be considered clinically actionable.

REGULATORY DISCLOSURES

This report was generated by a personal/educational bioinformatics pipeline (RareDiseasePipeline) built for pipeline development and validation purposes. It has not been developed, validated, or reviewed by any accredited clinical laboratory, has not been cleared or approved by the FDA, and is not a CLIA- or CAP-certified clinical test result. It must not be used for patient care, diagnosis, or any clinical decision-making.

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